Major task WG.

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for TE: $E_y = \begin{cases} A\cos K_x x & -w < x < w \\ Be^{-\gamma x} & x > w \end{cases}$

$$P_{core} = \frac{1}{2Z_{gTE}} * \int_{-w}^{w} A^2 \cos^2 K_x x \, dx \quad \text{"even TE"}$$

$$P_{core} = \frac{1}{2Z_{gTE}} \times A^2 \times \frac{1}{2} \int_{-w}^{w} 1 + \cos(2K_x x) \, dx.$$

$$P_{core} = \frac{2}{2Z_{gTE}} \times \frac{A^2}{2} \int_{0}^{w} (1 + \cos 2K_x x) \, dx$$

$$P_{core} = \frac{A^2}{2Z_{gTE}} \left[w + \frac{\sin(2K_x w)}{2K_x} \right]$$

$$P_{clad} = \frac{2}{2Z_{gTE}} * \int_{+w}^{\infty} B^2 e^{-2\gamma x} \, dx = \frac{B^2}{Z_{gTE}} \left[\frac{e^{-2\gamma x}}{-2\gamma} \right]_{w}^{\infty}$$

$$P_{clad} = \frac{B^2}{2Z_{gTE}} \left[\frac{e^{-2\gamma w}}{\gamma} \right]$$

$$\Gamma_{TE} = \frac{A^2 \left(w + \frac{1}{2K_x} \sin(2K_x w) \right)}{\frac{B^2 e^{-2\gamma x}}{\gamma} + A^2 \left(w + \frac{1}{2K_x} \sin 2K_x w \right)} \quad (1)$$

from B.Cs $A\cos K_T w = Be^{-\gamma w} \rightarrow \frac{A^2}{B^2} = \frac{e^{-2\gamma w}}{\cos^2 K_x w}$ (2)

- Dispersion Relation

$$K_x \tan K_x w = \gamma \quad (3)$$

$$\gamma w = V\sqrt{b} \;, \quad K_x w = V\sqrt{1-b} \quad (4)$$

from (3) & (4) $\tan(K_x w) = \frac{\gamma}{K_x} = \frac{\gamma w}{K_x w} = \frac{V\sqrt{b}}{V\sqrt{1-b}} = \sqrt{\frac{b}{1-b}}$

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$\sin(k_x W) = \sqrt{b}$

$\cos(k_x W) = \sqrt{1-b}$

$$\sin 2k_x W = 2\sin(k_x W)\cos(k_x W)$$
$$= 2\sqrt{b}\sqrt{1-b}$$

$$\Gamma_{TE} = \frac{\left(W + \frac{1}{2k_x} 2\sqrt{b}\sqrt{1-b}\right)}{\frac{\cos^2 k_x W}{\gamma e^{-2\gamma x}} e^{-2\gamma x} + \left(W + \frac{1}{2k_x} 2\sqrt{b}\sqrt{1-b}\right)}$$

$$\Gamma_{TE} = \frac{\gamma W + \frac{\gamma W}{k_x W}\sqrt{b}\sqrt{1-b}}{\cos^2 k_x W + \gamma W + \frac{\gamma W}{k_x W}\sqrt{b}\sqrt{1-b}}$$

$$\Gamma_{TE} = \frac{V\sqrt{b} + \sqrt{\frac{b}{1-b}}\sqrt{b}\sqrt{1-b}}{1-b+V\sqrt{b} + \sqrt{\frac{b}{1-b}}\sqrt{b}\sqrt{1-b}} = \frac{V\sqrt{b}+b}{1-b+V\sqrt{b}+b}$$

$$\Gamma_{TE} = \frac{V+\sqrt{b}}{V+\frac{1}{\sqrt{b}}}$$

TE_{odd} :- $E_y = \begin{cases} A\sin k_x x \\ B e^{\gamma x} \end{cases}$ $\cot(k_x W) = \frac{-\gamma W}{k_x W}$

$$\frac{B}{A} = \frac{\sin k_x W}{e^{-\gamma W}}$$

$$\Gamma_{TE_{odd}} = \frac{\int_{-W}^{W} (A^2 \sin^2 k_x x)\, dx}{2\int_{W}^{\infty} B^2 e^{-2\gamma x} + \int_{-W}^{W} A^2 \sin^2(k_x x)\, dx}$$

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$$\Gamma_{TE_{odd}} = \frac{A^2\left[W - \frac{\sin 2K_x X}{2K_x}\right]}{\frac{B^2 e^{-2\gamma W}}{\gamma} + A^2\left[W - \frac{\sin(2K_x X)}{2K_x}\right]}$$

$$= \frac{A^2(K_x W - \cos K_x W \sin K_x W)}{A^2 \frac{\sin^2 K_x W}{e^{-2\gamma W}} e^{-2\gamma W} * \frac{K_x}{\gamma} + A^2\left[K_x W \cos(K_x W)\sin(K_x W)\right]}$$

$$\Gamma = \frac{V\sqrt{1-b} + \frac{V\sqrt{b}}{V} \cdot \frac{V\sqrt{1-b}}{V}}{\frac{V^2(1-b)}{V^2} \times \frac{\sqrt{1-b}}{\sqrt{b}} + V\sqrt{1-b} + \sqrt{b}\sqrt{1-b}} = \frac{V+\sqrt{b}}{V+\frac{1}{\sqrt{b}}}$$

$$\Gamma_{TE_{even}} = \Gamma_{TE_{odd}}$$

TM even: $H_y = \begin{cases} A\cos K_x X \\ B e^{-\gamma x} \end{cases}$, $E_x = \frac{B}{\epsilon_0 n_i} H_y$

$$\tan K_x W = \frac{\eta\gamma}{K_x}$$

$$\frac{B}{A} = \frac{\cos K_x W}{e^{-\gamma W}}$$

$$\Gamma = \frac{\int_{-W}^{W} \cos^2(K_x X)\,dx}{2\int_{W}^{\infty} \frac{B^2}{A^2} e^{-\gamma x}\,dx + \int_{-W}^{W} \cos^2(K_x X)\,dx}$$

$$= \frac{\left[W + \frac{\sin(2K_x W)}{2K_x}\right]\frac{1}{n_1^2}}{\frac{2}{n_2^2}\frac{\cos^2 K_x W}{e^{-2\gamma W}}\frac{e^{-2\gamma W}}{2\gamma} + \left[W + \frac{\sin 2K_x W}{2K_x}\right]\frac{1}{n_1^2}}$$

$$= \frac{V\sqrt{1-b} + \frac{V\eta\sqrt{b}\,V\sqrt{1-b}}{V^2\left[(\eta^2-1)b+1\right]}}{\eta \cdot \frac{V^2(1-b)}{V^2\left[(\eta^2-1)b+1\right]} \cdot \frac{\sqrt{1-b}}{\sqrt{b}} + V\sqrt{1-b} + \frac{\eta\sqrt{b}\sqrt{1-b}}{(\eta^2-1)b+1}}$$

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$$\Gamma_{TM\ even} = \frac{V((\eta^2-1)b+1) + \eta\sqrt{b}}{V((\eta^2-1)b+1) + \frac{\eta}{\sqrt{b}}}$$

TM_{odd}:

$$H_y = \begin{matrix} A\sin K_x w \\ Be^{-\gamma x} \\ -Be^{\gamma x} \end{matrix}, \quad E_x = Z_g^{TM} H_y = \frac{\beta}{\epsilon_0 n_1^2} H_y$$

$$\Gamma_{TM_{odd}} = \frac{\int_{-w}^{w} \frac{A^2}{n_1^2}\sin^2(K_x x)dx}{2\int_{w}^{\infty} \frac{B^2}{n_2^2} e^{-2\gamma x}dx + \int_{-w}^{w} \frac{A^2}{n_1^2}\sin^2(K_x x)dx} = \frac{P_{core}}{P_{clad}+P_{core}}$$

$$\Gamma_{TM} = \frac{\frac{A^2}{n_1^2}\left[w - \frac{\sin(2K_x x)}{2K_x}\right]}{\frac{B^2}{n_2^2}\frac{e^{-2\gamma w}}{\gamma} + \frac{A^2}{n_1^2}\left[w - \frac{\sin(2K_x w)}{2K_x}\right]}$$

$$\frac{B}{A} = \frac{\sin(K_x w)}{e^{-\gamma w}}, \quad \cot K_x w = \frac{-\eta\gamma w}{K_x w}$$

$$\Gamma_{TM_{odd}} = \frac{V\sqrt{1-b} + \frac{\eta\sqrt{b}\,V\sqrt{1-b}}{V^2((\eta^2-1)b+1)}}{\frac{\eta V^2(1-b)}{V^2((\eta^2-1)b+1)}\frac{\sqrt{1-b}}{\sqrt{b}} + V\sqrt{1-b} + \frac{\eta\sqrt{b}\sqrt{1-b}}{(\eta^2-1)b+1}}$$

$$\Gamma_{TM_{odd}} = \frac{V[(\eta^2-1)b+1] + \eta\sqrt{b}}{V[(\eta^2-1)b+1] + \eta/\sqrt{b}}$$

$$\Gamma_{TM_{even}} = \Gamma_{TM_{odd}}$$